DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 1A

Question booklet · 30 marks · 90 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator. Record choices online or on separate paper.

QUESTION 1 [1]

An insect lifts its foot from a pond surface. A detergent reduces the upward force at the same contact length. Which property of water best explains the original force?

A Water adhesion to the foot alone, independently of attraction between water molecules
B The density difference between liquid water and the air above it
C Cohesion among water molecules at the interface
D High specific heat capacity, which resists local warming by the insect

QUESTION 2 [1]

A double-stranded DNA fragment contains 180 base pairs, including 54 G–C pairs. How many thymine nucleotides does it contain?

A 126
B 72
C 54
D 252

QUESTION 3 [1]

A cell is 18 µm wide. Its image measures 7.2 mm on the screen. What is the linear magnification?

A ×0.4
B ×40
C ×2500
D ×400

QUESTION 4 [1]

Two isolated bird populations look alike. Which observation most directly supports treating them as separate biological species?

A They occupy forests on different islands
B When brought into contact they repeatedly fail to produce fertile offspring despite normal breeding conditions
C Their mean adult masses differ by 3%
D They have different frequencies of a plumage allele

QUESTION 5 [1]

A fossil mammal has a forelimb with the same sequence of bones as a bat wing, but different bone proportions. Which inference is best supported?

A The limbs may be homologous despite different functions
B The fossil must be a direct ancestor of bats
C The limbs must have evolved independently because their shapes differ
D The two species necessarily occupied identical niches

QUESTION 6 [1]

An island breeding programme can house only 20 founders. Which selection best protects genetic diversity across the source population?

A Choose individuals expressing one preferred colour
B Use ten sibling pairs that already breed successfully
C Choose the twenty heaviest adults from one family
D Sample minimally related individuals across the source range

QUESTION 7 [1]

Two polymers yield glucose on complete hydrolysis. Human amylase acts on one but not the other. Which difference best explains this?

A The arrangement of glucose is identical, but amylase recognizes the free chain ends only
B The orientation of the glycosidic linkages differs
C The number of glucose residues alone determines whether either polymer can bind amylase
D Cellulose is resistant because all bonds connecting its chains are covalent

QUESTION 8 [1]

After heating, a protein loses activity. Chemical analysis detects no change in amino acid sequence. Which conclusion is justified?

A A changed side-chain sequence is the only way to alter active-site shape
B Loss of activity demonstrates cleavage of the polypeptide backbone into separate chains
C The active site must remain functional as long as peptide bonds remain intact
D The functional three-dimensional conformation may have changed

QUESTION 9 [1]

Cells equilibrate with a dye through a specific channel. ATP production is inhibited but a dye concentration gradient remains. What should occur while the channel stays open?

A Net movement against the gradient using the channel alone
B Movement stops because all membrane transport requires ATP
C Dye moves only after vesicles fuse with the membrane
D Net movement down the dye concentration gradient

QUESTION 10 [1]

A gland cell secretes a polypeptide enzyme. Which route would a newly made enzyme normally follow?

A Nucleus → lysosome → chloroplast
B Rough ER → Golgi apparatus → secretory vesicle
C Smooth ER → mitochondrion → nucleus
D Golgi apparatus → nucleus → rough ER

QUESTION 11 [1]

Two similarly shaped cells have widths of 10 and 20 μm. The larger cell has how much membrane area per unit volume compared with the smaller cell?

A Twice as much
B The same amount
C One half as much
D Four times as much

QUESTION 12 [1]

Fibrosis doubles the diffusion distance across a gas exchange surface. All other factors remain fixed. How does the initial oxygen diffusion rate change under an inverse-distance model?

A It is unchanged
B It falls to one quarter
C It halves
D It doubles

QUESTION 13 [1]

In a flow model a flexible one-way valve permits blood to pass from a chamber (left ventricle) into an artery (aorta), but prevents backflow. As pressure in the chamber rises from below to above the pressure in the artery, which statement describes the valve opening?

A Atrial pressure exceeds venous pressure and blood enters the aorta directly
B Equal pressure across the valve actively pumps blood forward
C Ventricular pressure exceeds aortic pressure and blood moves into the aorta
D Aortic pressure exceeds ventricular pressure and blood moves into the ventricle

QUESTION 14 [1]

Two desert plants have equal leaf area. Plant X has stomata recessed into still-air pockets. Which consequence is most likely in dry wind?

A X can take up carbon dioxide only at night
B A more humid local boundary layer reduces the water-vapour gradient
C X no longer requires xylem
D Water leaves faster because the diffusion path shortens

QUESTION 15 [1]

A grazer usually occupies an entire shore when its competitor is excluded. With the competitor present it occupies only the lower shore. What do the results suggest?

A Both species occupy identical realized niches everywhere
B Competition expands the fundamental niche
C The upper shore is always physiologically intolerable
D Competition restricts the realized niche

QUESTION 16 [1]

A substrate disappears from 8.0 to 6.2 mmol in 30 s. A matched tube lacking enzyme loses 0.30 mmol. What is the background-corrected mean enzymatic rate?

A 0.070 mmol s^{-1}
B 0.050 mmol s^{-1}
C 0.010 mmol s^{-1}
D 0.060 mmol s^{-1}

QUESTION 17 [1]

Germinating seeds warm an insulated flask while an equal mass of boiled seeds does not. Which explanation best connects the observation to respiration?

A All energy from substrates is stored indefinitely as ATP
B Heat is produced only by photosynthesis
C Some chemical energy released from substrates is transferred as heat
D Oxygen supplies the carbon atoms of respiratory substrates

QUESTION 18 [1]

An illuminated algal culture releases oxygen at a net rate of 8 units per hour and consumes oxygen in darkness at 3 units per hour. Assuming the same respiration rate in light, what is its gross photosynthetic oxygen production?

A About 2.7 units per hour
B 5 units per hour
C 8 units per hour
D 11 units per hour

QUESTION 19 [1]

A sensory neuron carries an impulse from a receptor toward the spinal cord. Which change moves along its axon?

A Movement of the receptor cell to the spinal cord
B A wave of membrane depolarization
C Bulk flow of neurotransmitter inside the axon
D Diffusion of an action potential through blood

QUESTION 20 [1]

A bright light reaches one eye and its pupil becomes smaller. Which sequence best represents the reflex pathway?

A Receptor → sensory neuron → central integration → motor neuron → effector
B Receptor → motor neuron → sensory neuron → effector
C Central integration → receptor → sensory neuron → effector
D Effector → motor neuron → receptor → sensory neuron

QUESTION 21 [1]

After a booster dose, antibody concentration reaches a higher peak sooner. Which mechanism best explains this?

A Antigen-specific plasma cells arise from erythrocytes during the booster response
B Primary antibodies cause all later antigens to be removed before recognition
C Memory lymphocytes initiate a rapid specific response
D Memory B cells produce a faster response by lowering the specificity of antigen recognition

QUESTION 22 [1]

A closed population of beetles has 80 marked adults. A later unbiased sample contains 12 marked and 48 unmarked adults. What population estimate follows from equal capture probabilities and thorough mixing?

A 320
B 400
C 100
D 480

QUESTION 23 [1]

Primary consumers ingest 600 kJ from plants and add 54 kJ to their own biomass. What percentage of ingested energy becomes new consumer biomass?

A 90%
B 54%
C 11.1%
D 9%

QUESTION 24 [1]

Both parental DNA strands carry a heavy label. Replication occurs twice in light medium. What fraction of double-stranded molecules still contains a heavy strand?

A One half
B One quarter
C None of them
D All of them

QUESTION 25 [1]

A coding DNA strand reads 5′-ATG GAA TTT-3′. Which mRNA sequence is transcribed from the complementary template?

A 5′-AUG GAA UUU-3′
B 5′-ATG GAA TTT-3′
C 5′-AAA UUC CAU-3′
D 5′-UAC CUU AAA-3′

QUESTION 26 [1]

A substitution changes a coding triplet but leaves the amino acid unchanged. Which statement explains the result?

A The ribosome reads overlapping triplets and thereby corrects a changed base
B Different codons can specify the same amino acid
C A substitution changes mRNA but can never affect the encoded amino acid
D All tRNAs carry more than one amino acid at a time

QUESTION 27 [1]

A diploid cell contains 12 chromosomes in G1. Immediately after DNA replication and before mitosis, how many chromosomes and chromatids are present?

A 24 chromosomes and 24 chromatids
B 6 chromosomes and 12 chromatids
C 12 chromosomes and 24 chromatids
D 12 chromosomes and 12 chromatids

QUESTION 28 [1]

Plant tissue gains 6% mass in solution X and loses 3% in solution Y. Samples began in the same condition. Which inference is supported?

A X favours net water entry and Y favours net water loss
B The tissue water concentration was identical to both solutions
C Both solutions are isotonic with the tissue
D Solute must have moved by active transport in both

QUESTION 29 [1]

A heterozygote with a dominant phenotype is crossed with a homozygous recessive individual. What is the probability that two successive offspring both show the recessive phenotype?

A ¼
B ½
C ¾
D ⅛

QUESTION 30 [1]

Dark beetles survive better on dark ground. After several generations their frequency increases. What further observation most directly connects this to natural selection?

A Colour is heritable and dark survivors leave more surviving offspring

B Every beetle has an equal number of offspring

C Predators always prefer the most abundant beetle

D Individual beetles darken whenever predators appear